

HARSHIT DEV YADAV

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Work Experience

Software Engineer | Tohans Pvt Ltd, Bangalore

Jul'24 - Present

- Owned product surface of a smart POS calculator from 0 → **6,000+ devices in 14 months** — now the **company's primary revenue line**. Translated **merchant pain points** from on-ground visits and call logs into firmware features prioritized with founders against a **6-week roadmap**.
- Designed merchant **billing flow with sub-second SKU search across 15,000+ products**; reduced perceived checkout time and lifted POS session completion (tracked in **Mixpanel**).
- Diagnosed manufacturing **bottleneck** capping throughput; **built 9-test automated factory diagnostic suite** — **lifted first-pass yield to 90%, cut per-device test time 60%**, reduced per-unit cost at scale.
- Drove end-to-end integration of **thermal printer and barcode scanner** — kernel drivers through merchant UI — coordinating with **mechanical and ops teams** for production rollout.

Software Engineer Intern | Upliance.ai, Bangalore

Mar'24 - Jul'24

- Shipped firmware for **precise temperature control** on a consumer kitchen device; designed the **safety state machine** on ESP32, owning **hardware-firmware integration end-to-end**.
- Operated in **pre-launch ambiguity** alongside founders — translated cooking-experience goals into firmware constraints (**sampling rate, fail-safe defaults**) that directly shaped **UX and unit economics**.

Skills

AI & Prototyping: Claude API, OpenAI API, prompt engineering, agentic workflow design, vibe coding (Claude code, Codex)

Product & Analytics: SQL, Python, Mixpanel, Jira, Figma, Sentry, Excel

Engineering Depth: C/C++17, CMake, GDB, REST APIs, Gtest, GitHub

Education

Lovely Professional University, India

2020 - 2024

- Bachelor of Technology (B.Tech) in Computer Science

Projects

DocPilot [MVP](#) | [Github](#)

Apr'26

AI doc agent for NBFC loan officers that automates WhatsApp doc collection, OCR to CRM auto-fill

- Workflow scoping** - one painful loop over one channel; agent chases customers for bad/missing docs and escalates edge cases to officer only.
- Human-in-the-loop** by architecture, officer gets one-click approve/reject only when human judgment is needed.
- 4 real applicant scenarios** in demo (clean run, retake, chases, escalation) to surface failure modes. Stack: React 19, Vite, Tailwind.

Loan EMI Negotiator [MVP](#) | [Github](#)

May'26

AI agent generating lender-specific rate-negotiation artifacts for Indian borrowers, closes the gap between informing and acting.

- Product wedge:** 3 scope decisions: one painful workflow over generic AI finance; action artifact (editable email + script) over chat; trust design (citations, no auto-send) over raw model quality.
- Reliability as a product feature:** 4-tier Gemini model fallback ladder with deterministic demo mode and X-Demo-Reason audit headers, never shows a broken state during live demo or quota spike.
- Stack:** Next.js 16, TypeScript, AI SDK + Gemini, Zod, Tailwind, ~90s end-to-end generation.

Open Source Contributions

Meta OpenZL - Format-Aware Compression Framework | **C++17, Graph Algorithms** |

Contributor | C++, Graph-Based Compression

- Implemented N-to-N graph routing feature** enabling parallel stream processing through distinct compression pipelines, merged into Meta's production compression framework
- Code reviewed and approved by **Yann Collet** (creator of Zstandard), deployed to Meta's internal systems
- GitHub: facebook/openzl | **PRs:** [#229](#), [#220](#)

Academic and Extracurricular Achievements

- Top 200** Teams out of **8000** teams, **Flipkart Grid Hackathon** (Health Track) 2023
- Top 1000** People out of **125000** participants in **EY Techathon 4.0**, 2023